

A VLM ANSWER IS NOT AN ANOMALY SCORE: RANK COMPRESSION IN TRAINING-FREE VIDEO ANOMALY DETECTION

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ABSTRACT

Vision–language models enable training-free video anomaly detection by answering questions about video segments. VAD benchmarks, however, require a scalar anomaly score for each segment and evaluate the resulting ranking using the AUROC or AP. A VLM-based detector should therefore define an **answer interface**: the answer scale specifies the admissible answers, and the readout rule maps the model’s output distribution to a score. Because this interface can change the evaluated ranking, it is part of the detector rather than a formatting detail. The generated readout uses only the most likely answer, whereas the probability readout uses the full distribution over admissible answers. Across four 7–8B VLMs, the probability readout outperforms the generated readout for every tested combination of answer scale, benchmark, and metric, with average gains ranging from 5 to 13 points across the four benchmark–metric pairs. The gap arises because the generated readout keeps only one answer value per segment, so segment with different answer distributions can receive the same score and lose their relative order. We call this loss of relative order **generated-answer rank compression**. Even when the answer scale allows 91 answers, the generated readout produces only 4–18 distinct scores, whereas the probability readout retains substantially finer score resolution. The advantage persists under every decoding strategy, prompt wording, and joint scoring–explanation prompt we test. The answer interface is therefore a consequential component of VLM-based VAD and should be explicitly specified and evaluated.

1 INTRODUCTION

Video anomaly detection (VAD) temporally localizes events that depart from expected behavior. Anomalies are rare, open-ended, and strongly dependent on scene context, so conventional systems train task-specific detectors under semi-supervised or weakly-supervised settings (Sultani et al., 2018; Tian et al., 2021). Vision–language models (VLMs) offer a different route. Their broad visual and semantic priors support open-vocabulary recognition (Wu et al., 2024a), language-mediated anomaly scoring (Zanella et al., 2024), and user-defined anomaly descriptions (Liu et al., 2026b) without training a detector for deployment domain. A common generative VAD pipeline divides a video into temporal segments, prompts a frozen VLM to judge each segment, and converts each response into a scalar anomaly score. This conversion may seem like a minor implementation detail. The model can answer “Yes” or “No,” choose an integer from a bounded scale, or produce a floating-point rating. Alternatively, the detector can read the model probabilities of the admissible answers. All of these appear to be different ways of expressing the same underlying judgment. However, VAD benchmarks evaluate the ordering of the resulting scores, so these choices can change which segments are ranked above others.

AUROC and AP operate on scalar score rankings, whereas generation returns one string drawn from a distribution over the admissible answers. Suppose that distribution places 0.56 of its mass on “Yes” for one segment and 0.94 for another. Decoding the answer maps both to “Yes” and ties them, whereas reading the mass on “Yes” keeps the second segment ahead of the first. The visual input,

model parameters, anomaly definition, and temporal sampling are unchanged. Only the **answer interface** between language generation and ranking differs.

What information is lost when a decoded VLM answer becomes an anomaly score?

We study this interface through two controlled design axes. The **answer scale** fixes what the model may say: verbal binary, numeric binary, or floating-point. The **readout rule** fixes how what it says becomes a score. GEN keeps only the value of the most likely answer, whereas PROB derives the score from the full answer distribution. Reading a distribution rather than a decoded label is established practice in LLM-based evaluation and document ranking (Liu et al., 2023; Zhuang et al., 2024), but generative VAD systems have typically read out the answer. Under GEN, segments whose answer distributions differ receive the same score whenever they share a most likely answer. We call this loss of relative order **generated-answer rank compression**. PROB, in contrast, preserves this variation and can therefore distinguish segments that share the same most likely answer.

PROB wins every model–scale–dataset–metric comparison across four open 7–8B VLMs and seven answer scales, by an average of 7.66 AUROC and 12.75 AP points on UCF-Crime (Sultani et al., 2018) and 5.09 AUROC and 10.97 AP points on XD-Violence (Wu et al., 2020). The gap is caused by score ties introduced during decoding. Resolving only the score ties left by GEN recovers 93–95% of the gap. A finer grid does not remove the ties, since even floating-point prompts admitting up to 91 answers realize only 4–18 distinct values (Figure 1). These ties also make generated-score AP depend on how equal scores are ordered, with a range exceeding 42 points for GEN. The advantage persists under every question wording and decoding strategy we test, on average at every temporal smoothing width, and when the model also explains its score. Both readouts come from the same forward pass, so the remedy adds no inference cost. Our contributions are fourfold:

- We formulate the answer-to-score conversion as an explicit design layer, factorized into answer scale and readout rule. We identify generated-answer rank compression as its central failure mode.
- We provide a controlled characterization of the interface across four VLMs, two benchmarks, and seven answer scales, showing that PROB wins every paired comparison, with paired video-level bootstrap intervals that exclude zero for every per-model and per-scale mean (Section 5.1).
- We verify the mechanism. Tie-breaking alone recovers 93–95% of the gap, finer scales relieve only GEN without removing the ties, and the same ties leave generated-score AP underdetermined by more than 42 points (Section 5.2 and Section 5.3).
- We show that the advantage survives under temporal smoothing, joint scoring–explanation prompts, question paraphrases, a polarity reversal, and sampling-based decoding, and that it shrinks materially only when the explanation is decoded ahead of the score (Section 5.4).

2 RELATED WORK

Task-specific video anomaly detection. Classical VAD learns a scoring function from video, under semi-supervised (Hasan et al., 2016; Gong et al., 2019; Liu et al., 2018) or weakly supervised (Sultani et al., 2018; Tian et al., 2021; Feng et al., 2021; Song et al., 2025) training. Across this literature, methods vary the supervision regime and the input representation, using signals such as RGB, optical flow, pose, and object trajectories (Hirschorn & Avidan, 2023; Song & Lee, 2025; 2026a). These lines vary the training signal, or the representation, and we vary none of them. Recent analyses question whether standard annotations and metrics capture deployed practice (Liu et al., 2025; Song & Lee, 2026b), and our tie-sensitivity results extend that critique to how a generative answer becomes

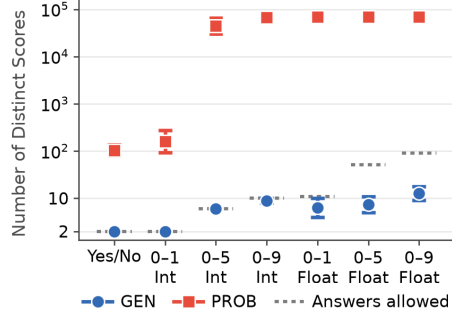


Figure 1: A decoded answer cannot carry the resolution the prompt asks for. On UCF-Crime the generated readout (GEN) never exceeds the number of admissible answers, and it realizes only about 13 distinct scores where 91 are allowed. The probability readout (PROB) is not bounded by that grid, and two admissible answers already yield more distinct scores than the finest grid does. Markers are the four-model mean and bars span the models.

a score. The step that turns a frozen generative detector’s judgment into a ranked score sequence can alter the result by itself.

Vision-language models for VAD. Language-based detectors divide on where scoring happens and on what the model is asked to emit. CLIP-based methods keep scoring outside generation, aligning frame features with semantic anomaly prompts under weakly supervised or open-vocabulary training (Radford et al., 2021; Wu et al., 2024b;a; Xu et al., 2025; Zhu et al., 2026; Li et al., 2025). Generative systems move scoring into the language output, and they then differ in the answer they request. Some request a bounded rating on a fixed grid, alone or attached to a short justification (Zanella et al., 2024; Ahn et al., 2026; Song et al., 2024), while others organize the judgment around rules, verbalized explanations, or multi-granularity reasoning (Yang et al., 2024; Ye et al., 2025; Zhang et al., 2025a). Whatever is requested, the score is the answer the model produced, with ESOM the closest to our position in mapping generated intervals and token-level confidence to soft frame scores for streaming (Liu et al., 2026a). Downstream refinement is just as common and just as system-specific, spanning similarity-weighted aggregation over related segments, fusion of keyword-conditioned score streams, and near-universal temporal smoothing (Zanella et al., 2024; Ye et al., 2025; Ahn et al., 2026). We hold every stage before the answer fixed and vary only the reduction that follows.

LLMs as probabilistic judges and rankers. Outside VAD, next-token likelihoods already serve as task scores. G-Eval diagnoses the same compression in text evaluation, where an integer-decoded quality score assigns most candidates the same value, and repairs it with a probability-weighted sum over the admissible scores (Liu et al., 2023). Pointwise rankers normalize relevance-label likelihoods to order documents, where graded labels can improve discrimination while ranking by generated labels yields extensive ties and lower retrieval performance (Zhuang et al., 2024). Generative verifiers score correctness by the probability of a “Yes” token, optionally after a verification rationale (Zhang et al., 2025b). What these studies establish is the repair. What they cannot supply is its consequences in temporal detection, where three of their extrapolations fail. Graded labels improve generated ranking in retrieval, yet a finer scale buys PROB nothing here and a binary interface is often its best. Document ties are exchangeable, whereas VAD’s concentrate at the top of a temporally structured ranking, where AP prices them hardest. Neither task post-processes its scores, whereas VAD smooths by default, which partially conceals the loss. Throughout, we use answer probabilities as a relative ranking signal over an admissible set rather than as calibrated probabilities, since generative models are frequently miscalibrated (Guo et al., 2017; Kadavath et al., 2022).

3 THE ANSWER INTERFACE

One forward pass over a temporal unit yields a distribution over the admissible answers of the prompted scale. Everything studied in this paper happens after that distribution exists. A readout rule collapses it to a scalar, either by taking the value of the most likely answer or by taking its expected value under the distribution, and the resulting sequence of scalars is what the benchmark ranks. What pipelines describe as score extraction is therefore a choice between reductions, and this section makes the choice set explicit. We first fix the scales and the notation, then define the two readouts, then show why the first is a many-to-one map and what that map costs each ranking metric.

3.1 PROBLEM SETUP AND ANSWER SCALES

Let x_t be the visual input of temporal unit t and q_k an anomaly query whose requested response follows scale k . A frozen VLM defines a conditional distribution $p_\theta(y \mid x_t, q_k)$ over textual answers y . An answer interface $\mathcal{I} = (k, r)$ pairs scale k with a readout rule $r \in \{\text{GEN}, \text{PROB}\}$ and maps the model output to a normalized anomaly score

$$s_t^{(k,r)} = \mathcal{R}_{k,r}[p_\theta(\cdot \mid x_t, q_k)] \in [0, 1], \quad (1)$$

where $\mathcal{R}_{k,r}$ is the readout map instantiated in Section 3.2 and Section 3.3. With model and unit fixed, the score varies only through scale and readout. Each scale k fixes a finite ordinal answer set $\mathcal{A}_k = \{a_0, \dots, a_K\}$ whose $K+1$ admissible answers map to evenly spaced values $v_k(a_i) = i/K$ in $[0, 1]$. We instantiate verbal binary (Yes/No), numeric binary (0–1), and ordinal 0–5 and 0–9 scales.

Three floating-point scales request one decimal place over 0–1, 0–5, and 0–9, giving finite answer sets of 11, 51, and 91 values. Normalizing all values to $[0, 1]$ separates granularity from numeric range. Every prompt shares the same anomaly semantics, differing only in the rating clause and response format (Section D).

3.2 GENERATED READOUT

Let $\ell_t(a) = \log p_\theta(a \mid x_t, q_k)$ be the sequence log-likelihood the VLM assigns to answer $a \in \mathcal{A}_k$ after the prompt. GEN keeps only the most likely admissible answer:

$$y_t^* = \arg \max_{a \in \mathcal{A}_k} \ell_t(a), \quad s_{t,\text{GEN}}^{(k)} = v_k(y_t^*). \quad (2)$$

Equation (2) is an idealized greedy decoder with a fully compliant parser. The selected answer is always valid, so any gap to PROB is attributable to the reduction, not to formatting or parsing failures that unconstrained generation would add. On Yes/No, the idealization is measured rather than assumed. The models tested place essentially all of their probability on the admissible answers, and the unconstrained greedy token equals the constrained argmax on every scored unit (Section F). By construction, the GEN sequence contains at most $|\mathcal{A}_k|$ distinct values.

3.3 PROBABILITY READOUT

PROB retains the entire answer set. It normalizes the answer likelihoods within the scale,

$$\tilde{p}_t(a) = \frac{\exp \ell_t(a)}{\sum_{a' \in \mathcal{A}_k} \exp \ell_t(a')}, \quad (3)$$

and scores each unit by the expected normalized answer value,

$$s_{t,\text{PROB}}^{(k)} = \sum_{a \in \mathcal{A}_k} v_k(a) \tilde{p}_t(a). \quad (4)$$

For Yes/No, this reduces to the normalized probability mass on “Yes.” For multi-token floating-point answers, $\ell_t(a)$ is the full-sequence token log-likelihood of the numeric string, and the treatment of decimal-point and digit tokens is immaterial to either readout (Section S). Both readouts come from one forward pass, so a paired comparison shares the video units, anomaly definition, model parameters, inference settings, and answer likelihoods. Only the final reduction differs.

3.4 GENERATED-ANSWER RANK COMPRESSION

GEN is a many-to-one map from answer distributions to answer values. When different distributions yield the same top-ranked answer, it removes their relative ordering and increases ties. Let s^+ and s^- be scores of randomly sampled anomalous and normal frames. AUROC admits the ranking form

$$\text{AUROC}(s) = \Pr(s^+ > s^-) + \frac{1}{2} \Pr(s^+ = s^-), \quad (5)$$

so ties earn half the credit of correct orderings. A finer scale adds levels and can mitigate the tie growth. PROB instead keeps an ordering inside a fixed answer, so it can order frames that receive the same answer even when only two answers are admissible. Average precision is not expressed by Equation (5). Let $v_1 > \dots > v_K$ be the distinct scores, p_k and n_k the anomalous and normal counts at v_k , and TP_k, FP_k their cumulative sums over $j \leq k$. Evaluating each distinct score once,

$$\text{AP}(s) = \sum_{k=1}^K \frac{p_k}{P} \cdot \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k}, \quad P = \text{TP}_K. \quad (6)$$

Both metrics are fixed by the ranked list and its ties, but they price a tie differently. Equation (5) charges a tied pair one half wherever it falls in the ranking, whereas in Equation (6) the p_k anomalous frames at v_k share a single precision whose denominator carries their own n_k normal frames. That denominator is small near the top, so the same tie costs more the higher it sits. Compression should therefore cost more under AP than under AUROC (verified in Section 5.2).

Table 1: PROB outperforms GEN in all 28 scale-metric cells, most on the binary scales (mean over four VLMs, percentage points). Δ is the paired PROB-GEN gain. Per-model results are in Table 3 and Table 4.

Scale	UCF-Crime						XD-Violence					
	AUROC			AP			AUROC			AP		
	GEN	PROB	Δ	GEN	PROB	Δ	GEN	PROB	Δ	GEN	PROB	Δ
Yes/No	72.20	85.32	+13.12	20.96	39.88	+18.91	79.80	90.80	+11.00	49.44	68.98	+19.54
0-1 int	73.69	85.42	+11.73	21.39	39.50	+18.11	82.73	90.71	+7.98	50.53	68.08	+17.55
0-5 int	79.05	84.76	+5.71	28.18	39.67	+11.49	86.93	90.27	+3.34	57.58	66.96	+9.38
0-9 int	78.71	84.38	+5.67	26.77	38.26	+11.48	87.20	90.25	+3.06	58.71	66.94	+8.23
0-1 float	74.91	83.68	+8.78	27.16	38.16	+11.00	84.39	89.81	+5.43	57.32	65.67	+8.34
0-5 float	80.37	84.52	+4.14	30.10	39.77	+9.67	87.59	90.07	+2.48	59.41	66.54	+7.12
0-9 float	79.73	84.22	+4.49	30.07	38.65	+8.58	87.86	90.17	+2.31	60.22	66.85	+6.63

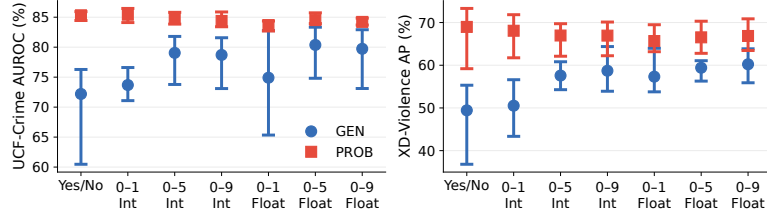


Figure 2: The same means as Table 1, on each benchmark’s primary metric. Markers are the four-model mean and bars span the model-wise range, not a confidence interval. Finer scales mainly benefit GEN, whereas PROB stays in a narrow band.

4 EXPERIMENTAL PROTOCOL

Models and benchmarks. We evaluate four frozen VLMs: Qwen3-VL-8B (Bai et al., 2025a), Qwen2.5-VL-7B (Bai et al., 2025b), InternVL3.5-8B (Zhu et al., 2025), and MiniCPM-V-4.5-8B (Yu et al., 2026). The benchmarks are the UCF-Crime test split of 290 videos and the XD-Violence test split of 800 videos (full dataset statistics in Section A).

Protocol and metrics. Following VERA (Ye et al., 2025), each video is divided into 16-frame units, and each unit is judged once from eight frames sampled uniformly over a 300-frame window centered on it (per-model input packaging in Section A). The unit score is broadcast to the unit’s frames, and we report frame-level AUROC and AP in percentage points, with AUROC primary on UCF-Crime and AP primary on XD-Violence. AP follows Equation (6), which is invariant to how frames sharing a score are ordered. Paired values throughout the paper report UCF-Crime first and XD-Violence second. For smoothing, each video’s unit-score sequence is convolved with a Gaussian kernel of width $\sigma \in \{1, 2, 4, 8, 10, 16, 20, 24\}$ before frame expansion. The eight widths and the unsmoothed scores give nine matched evaluation settings. Unsmoothed scores diagnose the answer interface directly, whereas smoothed scores test whether its effect survives matched post-processing.

5 RESULTS

5.1 PROBABILITY READOUT IMPROVES EVERY COMPARISON

PROB outperforms GEN for every model, scale, dataset, and metric, by a mean primary-metric gain of +7.66 AUROC points on UCF-Crime and +10.97 AP points on XD-Violence (Table 1, with per-model results in Table 3 and Table 4). The gains are not sampling noise, since the 95% interval of a paired video-level bootstrap excludes zero for every per-model and per-scale mean (Section C). The readout can matter more than the model, since under Yes/No on UCF-Crime the worst PROB model (84.45) exceeds the best GEN model (76.28). Model size does not close the gap either, and scaling two of the models to 32B and 38B widens it instead (Section L).

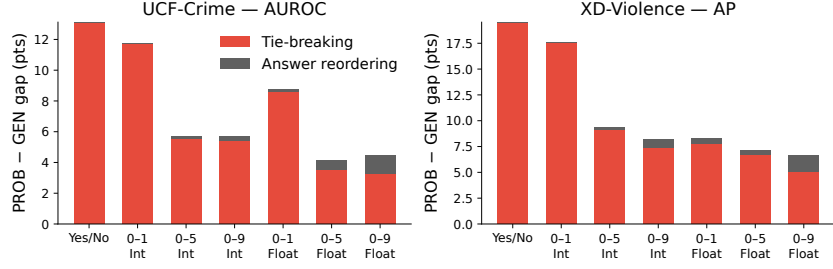


Figure 3: Decomposition of the PROB-GEN gap on each benchmark’s primary metric, averaged over four VLMs. Breaking ties among equal generated scores with PROB reproduces nearly the entire gap, and answer reordering contributes little outside the finest scales. Both metrics and exact values are in Table 10.

The gap is wider under AP than under AUROC in 53 of the 56 model-scale-dataset cells, at about $1.7\times$ and $2.2\times$ the AUROC gain on the two benchmarks (Table 1). Section 5.2 traces the asymmetry to where the ties sit.

Binary interfaces produce the largest gaps and fine ordinal and floating-point scales smaller ones (Figure 7). On UCF-Crime AUROC, per-model gains span 0.99 to 25.17 points, from Qwen3-VL under the 0-9 floating-point scale to Qwen2.5-VL under Yes/No.

5.2 TIE-BREAKING EXPLAINS THE GAP AND ITS METRIC ASYMMETRY

Every frame carries both a generated and a probability score from the same forward pass, and many frames share the same generated score. We therefore compare three rankings: GEN, PROB, and a *tie-broken* GEN that keeps the generated scores and uses PROB only to order the frames whose generated scores are equal. Tie-broken GEN minus GEN measures the ordering lost to ties, and PROB minus tie-broken GEN measures the remaining disagreement over answer order.

Tie-breaking alone accounts for 93–95% of the mean gap on both datasets and both metrics (Figure 3). On the two binary scales tie-breaking is the entire gap by construction, since GEN there is a monotone coarsening of PROB. The informative cells are the five finer scales, where tie-breaking still accounts for 88–92% of the pooled gap, and for 67–75% under the 0-9 floating-point prompt alone, where the readouts genuinely disagree about order (Section J). The advantage is therefore not primarily that the expected answer value disagrees with the argmax about which answers rank higher. The argmax erases the ordering among frames that receive the same answer. Breaking the ties at random leaves AUROC unchanged and recovers under a fifth of the AP gap, so the rest requires the ordering the probabilities carry. A well-placed threshold does not close the gap either. Binarizing PROB at the test-set-optimal threshold upper-bounds every binary readout of its scores, and it still trails PROB by 7.0–7.6 points for every model on Yes/No UCF-Crime AUROC (Table 11).

The metric asymmetry predicted in Section 3.4 is realized. Perfect tie-breaking, which orders all anomalous frames first among the frames that share a generated score, is worth exactly $\frac{1}{2} \Pr(s^+ = s^-)$ AUROC points but $2.3\times$ and $3.0\times$ as many AP points on the two benchmarks (Section J). The ties sit at the top of the ranking, where the frames sharing the highest generated score are enriched in anomalies yet far from pure. The observed gaps, at $1.66\times$ and $2.16\times$, fall short of that ratio because PROB recovers a smaller share of the AP headroom than of the AUROC headroom. Partially ordering an impure top block already earns most of the available AUROC, while AP keeps charging for every normal frame left above an anomalous one. The wider AP gap is thus a property of where the ties occur, not of the readout favoring one metric.

The recovered ordering is worth detections at deployable false-alarm budgets. At a false-alarm rate of 0.01, PROB roughly doubles the anomalous frames detected on both benchmarks (16.7% against 8.6% on UCF-Crime), and ordering only the tied frames covers most of that distance (Section N).

Rank compression is also an AP reproducibility hazard. A generated score leaves the frames sharing it tied, and AP depends on how positives and negatives are arranged among them. Across the 28 model-scale configurations per dataset, the optimistic and pessimistic orderings consistent with GEN span more than 42 AP points on average, against about one point for PROB (Table 2). AUROC

Table 2: The ties leave GEN AP underdetermined by more than 42 points, PROB by about one (mean over 28 model–scale configurations, percentage points). *Negative-first* and *Positive-first* place negatives and positives first, respectively, among frames sharing a score. Grouped AP, the estimator used throughout the paper, treats each distinct score as one threshold. Width is Positive-first minus Negative-first. *Trapezoidal* integrates the precision–recall curve as the official XD-Violence evaluator does (Wu et al., 2020). It is invariant to how ties are ordered yet lands inside the tie interval, sitting 13–15 points above the grouped value for GEN through interpolation alone.

Dataset	Readout	Negative-first	Grouped	Trapezoidal	Positive-first	Width
UCF-Crime	GEN	18.94	26.38	41.28	62.74	43.79
	PROB	38.83	39.13	39.32	39.69	0.86
XD-Violence	GEN	42.51	56.17	69.13	85.44	42.94
	PROB	66.68	67.15	67.37	67.80	1.12

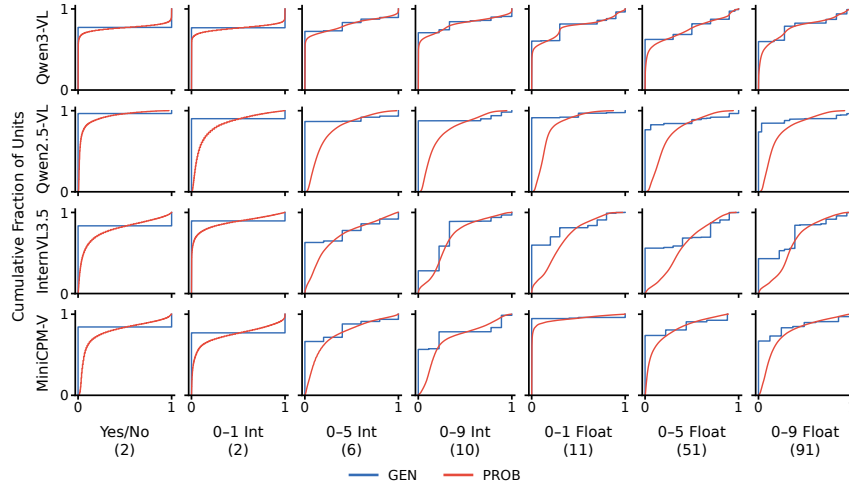


Figure 4: Empirical distribution of unit scores on UCF-Crime for every model and scale, with the admissible answer count in parentheses. GEN is an unbalanced staircase that widening the grid to 91 answers does not fix, whereas PROB rises almost continuously.

is immune because it assigns every tied positive–negative pair half credit. We therefore compute AP at distinct score thresholds (the non-interpolated estimator of Equation (6)) and report the tie interval separately (Section J). Integrating the precision–recall curve trapezoidally instead, as the official XD-Violence evaluation code do, adds 26.0 AP points on UCF-Crime and 20.7 on XD-Violence to GEN under the Yes/No interface while moving PROB by less than half a point (Section O). Reported AP can otherwise depend on an incidental sort order. An order-invariant grouped-threshold AP, together with reported score cardinality or tie headroom, removes that dependence.

The gain also extends to every anomaly type. All 13 UCF-Crime and all six XD-Violence classes improve on both metrics, with the largest AP gains on visually salient events and the smallest on subtle ones (Table 16 and Table 17).

5.3 A FINER SCALE DOES NOT SUBSTITUTE FOR THE READOUT

Scale moves GEN far more than PROB (Table 1 and Figure 2). Across models, GEN’s mean best-to-worst range spans 12–14 primary-metric points on the two benchmarks, while PROB’s spans 2–6. A fine scale is best for all eight GEN model–dataset pairs, but Yes/No or 0–1 is best for six of eight PROB pairs (Table 3 and Table 4). More levels relax quantization when a single decoded value is the score. Once within-answer probabilities preserve the ordering, extra verbalizers provide no systematic benefit.

The GEN half of that split is unanimous, while the PROB half holds only on average. At the mean over models, Yes/No PROB exceeds each of the five finer scales on both benchmarks, by up to

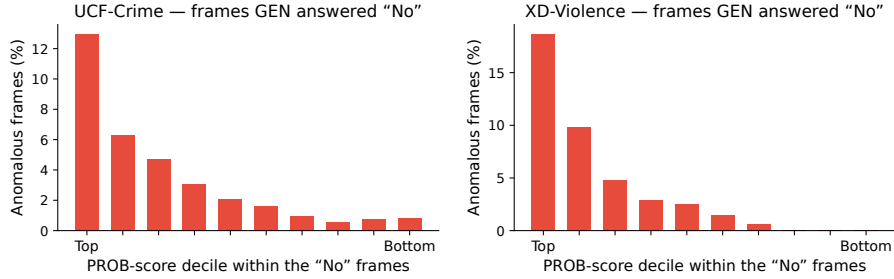


Figure 5: The ordering GEN discards inside a single answer finds real anomalies. We take every frame InternVL3.5-8B answers “No” under Yes/No, a group GEN scores identically, sort these frames by their PROB score, and split them into ten equal bins. The anomaly rate climbs toward the high-PROB bins, reaching $3.8\times$ (UCF-Crime) and $4.6\times$ (XD-Violence) the group’s average rate. All answers and models are reported in Table 12.

1.6 UCF-Crime AUROC and 3.3 XD-Violence AP points. The per-model preference behind the six-of-eight count is weaker, since three of the four UCF-Crime binary-versus-best-finer intervals cover zero and Qwen2.5-VL prefers a finer scale on XD-Violence AP (Section C). We therefore read the scale conclusion as a statement about the mean, not a per-model prescription.

Even for GEN, the grid delivers less than it admits (Figure 8 and Table 9). Integer GEN tracks its nominal 2, 6, or 10 levels, while floating-point GEN realizes at most 18 values despite admitting up to 91. The realized values are also unevenly occupied, with one score covering 72.4% and 55.1% of the units on average against under 1% for PROB. That score is 0.0 in 55 of the 56 configurations, so the grid is not merely coarse but piled onto “normal,” visible as the tall first step of every GEN curve in Figure 4. Under the same floating-point prompts, PROB assigns a distinct score to more than 99.8% of units.

The evidence hidden behind a fixed answer is real signal. Figure 5 shows what GEN discards inside a single answer. The frames InternVL answers “No”, 83.5% of UCF-Crime and 66.6% of XD-Violence, all share one generated score, so the anomalous frames among them can never be ranked above the normal ones. The figure sorts these frames by their PROB score, and the anomaly rate climbs toward the high-PROB bins. Over the whole group, PROB’s ordering reaches 76.9 and 84.2 AUROC (per-model ranges in Table 12). The ordering is not specific to one model, since ranking a model’s tied frames by a *different* model’s probabilities stays above chance in all 48 cross-model cells (Section M). The argmax discards detection signal, not noise.

Score cardinality is the symptom, not the cause. Binary PROB has only 79–273 distinct values (Section T), yet it is best in six of eight model–dataset pairs. Conversely, smoothed GEN scores are nearly continuous, yet the mean readout gap persists at every smoothing width (Figure 6). Few values in the right order therefore beat many values in the wrong one. Interpolation orders units by their neighbors, whereas PROB orders them by model evidence. Losing the within-answer order is the failure.

5.4 THE GAP SURVIVES SMOOTHING, DECODING, EXPLANATION, AND REPHRASING

Matched smoothing shrinks the mean gap to 1.92 and 1.87 points by $\sigma = 24$ (Figure 6 and Table 18). Gaussian smoothing (Zanella et al., 2024) interpolates a coarse sequence near transitions without recovering model probabilities. Yet PROB stays higher in 202 of the 252 matched pairs on UCF-Crime and 217 on XD-Violence (28 configurations across the nine settings), and granting each readout its own best σ , a test-set oracle that benefits GEN far more, leaves PROB ahead by 2.10 and 3.22 points.

Choosing the answer differently does not help. Compression follows from scoring a unit by a decoded answer, not from decoding the most likely one. On Yes/No, temperature sampling,

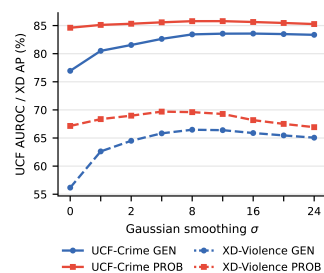


Figure 6: The mean readout gap narrows with σ but remains positive throughout. Bands span the 28 model–scale pairs.

nucleus sampling, and self-consistency by majority vote all stay on the same two scores and land at or below greedy decoding and far below PROB (Figure 9). Sixteen samples resolved by majority vote come within half a point of a single greedy decode. Averaging the samples rather than voting on them does escape the two-value set, since k samples admit $k+1$ scores, but it is then a Monte Carlo estimate of what PROB computes in closed form (Equation (4)). It approaches PROB from below, with one sample landing below greedy decoding, sixteen recovering 72% and 59% of the gap, and 256 still falling short of a single forward pass (Section K).

Generative detectors are used precisely because they can justify their judgments (Ahn et al., 2026; Ye et al., 2025; Zhang et al., 2025a). We therefore repeat the two binary-scale experiments with Qwen3-VL and MiniCPM-V, requesting a one-sentence scene description before or after the score. PROB exceeds GEN in all 16 primary-metric cells, by 8.29 and 17.72 points on average (Table 19), and in all 288 comparisons across metrics and smoothing settings (Section R). The readout effect is not an artifact of score-only prompting.

Output order matters selectively. Score-first PROB differs from score-only prompting by -0.90 points on average, while explanation-first PROB is lower in every cell (-3.39 on average) and GEN barely moves. Explanation-first decoding alters the generated answer on just 4.5–9.6% of scored units, yet it compresses the answer distribution. Entropy falls in all eight cells, and the share of ties PROB orders correctly falls in seven of the eight (Table 20). An explanation compresses the evidence behind a label it barely changes, a pattern consistent with rationale-then-verify readouts, which read an answer probability after a generated rationale (Zhang et al., 2025b). Under the interfaces we test, decoding the score before the explanation preserves the advantage.

The advantage also survives rephrasing. Three paraphrases of the Yes/No question leave PROB ahead of GEN in all eight model–wording cells, and PROB is the steadier readout under rewording (Section E). Reversing the question to ask about normality, which inverts which answer denotes an anomaly, leaves the gap essentially unchanged and bounds the model’s answer preference at 0.20 AUROC points (Section G). Section V places a single fixed interface, InternVL3.5-8B with Yes/No, among published VAD systems.

6 LIMITATIONS

The mean gain is unevenly carried. Per-model UCF-Crime AUROC gains range from 4.79 to 14.00 points against a mean of 7.66, with part of the skew traced to one model’s threshold placement (Table 11). The four models are a fixed set, not a random sample, so the intervals of Section C are conditional on them. Stating the effect for VLMs in general would require sampling checkpoints as a population instead of resampling videos.

The measured size is an unsmoothed, ranking-only quantity. The gap survives matched smoothing in aggregate, with the pooled interval excluding zero at $\sigma = 4$ and $\sigma = 8$. Individually, 16 and 17 of the 28 configurations do so at $\sigma = 4$, with seven of the eight binary interfaces among them. The oracle- σ residual carries no interval, because both σ are chosen on the test set. A deployment-relevant estimate would fix σ and the answer scale on a held-out split rather than on the evaluation split. Granting each readout its best answer scale shrinks the contrasts from 7.66 and 10.97 to 4.88 and 9.75 points. Throughout, the analysis concerns ranking quality, not calibration.

PROB requires exposed answer likelihoods. Open-weight models provide them, but many proprietary interfaces do not, and there the generated readout is the only option. In that setting our results do not offer a remedy, but they quantify what the decoded answer costs and which choices, a finer scale and matched smoothing, recover part of it.

Further limitations, on the confounded scale axis, the unmeasured mechanism, model and benchmark coverage, and unconstrained generation, are collected in Section U.

7 CONCLUSION

A VLM answer is not yet an anomaly score. We study how it becomes one. Decoding creates ties that finer prompts do not reliably prevent, and reading the answer probabilities instead avoids them. The ties account for 93–95% of the readout gap. The advantage survives matched smoothing and score-first explanation. The AP tie intervals expose the reproducibility cost of generated scores. Scale, readout, output order, tie policy, and smoothing jointly define a generative detector and should be reported with it. Choosing them deliberately turns the same frozen model into a stronger detector at no additional cost.

AI USE STATEMENT

In this work, we used generative AI tools to implement isolated utility functions in our model inference code, and these functions were verified and tested for correctness by the authors. We have not used generative AI tools for any other required-disclosure task. Additionally, we used generative AI tools for proofreading, grammar checking, and typographical checks during writing. We have reviewed all AI-assisted work, and every reported number was produced by the deterministic evaluation pipeline of the Reproducibility Statement. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work evaluates scoring rules on two public benchmarks, UCF-Crime and XD-Violence, using their standard test annotations, and redistributes no video data. The videos depict real events involving people, and we use them only to evaluate frame-level anomaly rankings. The scores and captions generated in our experiments are not used to identify, profile, or disparage any individual, and we will not release the generated captions. As with any anomaly detection method, deployment in surveillance settings requires safeguards against misuse.

REPRODUCIBILITY STATEMENT

Every result is produced by frozen public checkpoints and a deterministic readout, with no training and, outside the seeded decoding-strategy controls of Section 5.4, no sampling. The four models are Qwen/Qwen3-VL-8B-Instruct, Qwen/Qwen2.5-VL-7B-Instruct, OpenGVLab/InternVL3.5-8B and openbmb/MiniCPM-V-4.5, run in bfloat16. The temporal protocol is identical across models, using non-overlapping 16-frame units, judged from eight frames sampled at uniform spacing over a 300-frame window centred on the unit and clipped at video boundaries, with frame expansion back to the annotation grid before evaluation (Section A). Section D gives every prompt verbatim, including the paraphrases and the polarity reversal. The admissible answer strings are the labels of each scale and are tokenized without special tokens, so a candidate’s likelihood is the likelihood of exactly that string.

Both readouts are computed from one forward pass per unit, so GEN and PROB share the visual input, the model parameters and the answer likelihoods, and the paired comparisons are exact rather than matched. Evaluation uses the non-interpolated grouped-threshold estimator of Equation (6) and the tie-aware AUROC of Equation (5), and Section O states why the estimator choice matters and how much. Bootstrap results resample whole videos with a single replicate stream shared across readouts, models and scales, so every contrast drawn from them is paired.

The two knobs used by the controls are documented environment overrides rather than edits, namely inference dtype (Section T) and prompt variant (Section E and Section G). Code and the exact launch commands for every table will be released.

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A PROTOCOL DETAILS

Datasets. The UCF-Crime test split contains 290 videos, 1,111,808 scored frames, and 88,094 anomalous frames. The XD-Violence test split contains 800 videos, 2,335,801 scored frames, and 529,797 anomalous frames.

Temporal units and visual inputs. Each test video is divided into non-overlapping 16-frame temporal units, giving 69,634 scored units on UCF-Crime and 146,449 on XD-Violence. A unit is judged from eight frames sampled at uniform spacing over a 300-frame window centered on the unit and clipped at video boundaries. Qwen3-VL, Qwen2.5-VL, and InternVL3.5 receive the eight frames as an ordered image sequence in their chat templates. MiniCPM-V-4.5, whose public interface favors single images, receives the same frames composed into one 4×2 contact sheet. All models run frozen in bfloat16.

Shared computation of the two readouts. For every unit, a single prefill pass yields the log-likelihood of each admissible answer, and multi-token floating-point candidates are scored by teacher forcing over the numeric string. GEN and PROB are then computed from these same likelihoods, so each paired comparison in the study shares the visual input, the prompt, and all model computation, and differs only in the final reduction.

Smoothing and evaluation. Temporal smoothing convolves each video’s unit-score sequence with a Gaussian kernel truncated at 4σ with reflected boundaries. The (smoothed) unit score is repeated over the unit’s 16 frames, and frames are concatenated across videos before computing frame-level AUROC and AP. Tied pairs contribute one half to AUROC, as in Equation (5). AP is computed at distinct score thresholds so that frames sharing a score are evaluated jointly and the result is invariant to frame-array order.

B FULL PER-MODEL RESULTS

Table 3 and Table 4 give the complete results summarized in Table 1. Each comparison is paired at the forward-pass level, so GEN and PROB use the same answer likelihoods and differ only in their final reduction.

Table 3: Frame-level performance on UCF-Crime for every model and answer scale (percentage points; higher is better). PROB exceeds GEN in every cell. Underline marks the winner within each pair, and bold marks the winner in the Mean row.

Metric	Model	Yes/No		0–1 int		0–5 int		0–9 int		0–1 float		0–5 float		0–9 float	
		GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB
AUROC	Qwen3-VL-8B	76.28	<u>85.26</u>	75.23	<u>85.49</u>	78.93	<u>84.43</u>	78.97	<u>83.90</u>	82.76	<u>84.31</u>	82.85	<u>84.20</u>	82.91	<u>83.90</u>
	Qwen2.5-VL-7B	60.47	<u>85.64</u>	71.08	<u>85.61</u>	73.78	<u>85.81</u>	73.10	<u>85.87</u>	69.88	<u>82.70</u>	74.81	<u>84.20</u>	73.11	<u>84.42</u>
	InternVL3.5-8B	75.97	<u>85.93</u>	71.83	<u>86.46</u>	81.79	<u>84.96</u>	81.19	<u>84.34</u>	81.65	<u>84.37</u>	83.33	<u>85.73</u>	81.68	<u>84.85</u>
	MiniCPM-V-4.5-8B	76.07	<u>84.45</u>	76.60	<u>84.10</u>	81.70	<u>83.85</u>	81.57	<u>83.40</u>	65.34	<u>83.35</u>	80.50	<u>83.94</u>	81.23	<u>83.70</u>
	Mean	72.20	85.32	73.69	85.42	79.05	84.76	78.71	84.38	74.91	83.68	80.37	84.52	79.73	84.22
AP	Qwen3-VL-8B	19.91	<u>40.87</u>	18.98	<u>39.85</u>	26.47	<u>38.90</u>	27.12	<u>37.16</u>	30.52	<u>37.41</u>	31.49	<u>37.34</u>	31.86	<u>36.36</u>
	Qwen2.5-VL-7B	17.91	<u>37.46</u>	23.13	<u>39.44</u>	24.73	<u>39.79</u>	23.99	<u>39.04</u>	24.22	<u>38.46</u>	26.49	<u>39.43</u>	24.32	<u>39.11</u>
	InternVL3.5-8B	22.70	<u>39.39</u>	23.39	<u>39.00</u>	30.15	<u>39.67</u>	29.45	<u>38.38</u>	31.42	<u>39.67</u>	31.25	<u>41.52</u>	31.88	<u>39.72</u>
	MiniCPM-V-4.5-8B	23.34	<u>41.79</u>	20.08	<u>39.72</u>	31.37	<u>40.34</u>	26.54	<u>38.46</u>	22.46	<u>37.09</u>	31.15	<u>40.77</u>	32.24	<u>39.42</u>
	Mean	20.96	39.88	21.39	39.50	28.18	39.67	26.77	38.26	27.16	38.16	30.10	39.77	30.07	38.65

C VIDEO-LEVEL BOOTSTRAP

The main results contain more than three million scored frames, but frames from the same video are correlated. We therefore run a paired nonparametric bootstrap over every answer scale, that is, over 28 model–scale pairs per dataset. Each of 10,000 replicates resamples complete videos with replacement (290 on UCF-Crime, 800 on XD-Violence), and one resample per replicate is shared by both readouts, all four models, and all seven scales, so every gain stays paired. Frame-level metrics

Table 4: Frame-level performance on XD-Violence for every model and answer scale (percentage points; higher is better). PROB exceeds GEN in every cell. Underline marks the winner within each pair, and bold marks the winner in the Mean row.

Metric	Model	Yes/No		0-1 int		0-5 int		0-9 int		0-1 float		0-5 float		0-9 float	
		GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB
AUROC	Qwen3-VL-8B	84.64	<u>91.40</u>	80.35	<u>91.20</u>	86.78	<u>90.17</u>	87.16	<u>89.85</u>	88.03	<u>89.30</u>	88.74	<u>89.62</u>	88.82	<u>89.66</u>
	Qwen2.5-VL-7B	64.33	<u>88.62</u>	80.25	<u>88.87</u>	83.45	<u>88.50</u>	83.04	<u>88.70</u>	80.59	<u>88.43</u>	84.84	<u>88.36</u>	84.83	<u>88.85</u>
	InternVL3.5-8B	85.31	<u>91.67</u>	85.91	<u>91.59</u>	89.15	<u>91.76</u>	90.22	<u>91.73</u>	89.07	<u>91.35</u>	88.84	<u>91.45</u>	89.38	<u>91.44</u>
	MiniCPM-V-4.5-8B	84.92	<u>91.51</u>	84.42	<u>91.20</u>	88.33	<u>90.66</u>	88.35	<u>90.73</u>	79.85	<u>90.16</u>	87.93	<u>90.85</u>	88.41	<u>90.72</u>
	Mean	79.80	90.80	82.73	90.71	86.93	90.27	87.20	90.25	84.39	89.81	87.59	90.07	87.86	90.17
AP	Qwen3-VL-8B	50.66	<u>70.25</u>	43.35	<u>69.42</u>	54.60	<u>66.88</u>	55.66	<u>66.01</u>	57.64	<u>63.17</u>	59.42	<u>63.58</u>	59.84	<u>63.45</u>
	Qwen2.5-VL-7B	36.81	<u>59.19</u>	51.13	<u>61.75</u>	54.28	<u>62.11</u>	53.91	<u>62.22</u>	53.77	<u>63.22</u>	56.28	<u>62.79</u>	55.89	<u>64.07</u>
	InternVL3.5-8B	55.32	<u>73.31</u>	56.61	<u>69.31</u>	60.81	<u>69.13</u>	64.37	<u>70.12</u>	63.97	<u>69.49</u>	60.88	<u>69.46</u>	63.86	<u>70.87</u>
	MiniCPM-V-4.5-8B	54.97	<u>73.17</u>	51.01	<u>71.84</u>	60.64	<u>69.73</u>	60.91	<u>69.40</u>	53.92	<u>66.80</u>	61.06	<u>70.31</u>	61.30	<u>69.03</u>
	Mean	49.44	68.98	50.53	68.08	57.58	66.96	58.71	66.94	57.32	65.67	59.41	66.54	60.22	66.85

are then recomputed from the per-video score counts, with the same tie handling used throughout (Section A). All entries use unsmoothed scores, and before resampling every configuration reproduces its entry in Table 3 and Table 4. Table 5 reports percentile 95% intervals for the Yes/No interface and Table 6 for all seven scales. Each mean is formed within the shared replicate rather than by treating models or scales as independent samples.

Every per-model Yes/No interval and every per-scale mean interval excludes zero, on both benchmarks and both metrics. Individual model-scale pairs are less uniform. The interval excludes zero in all 28 XD-Violence pairs on both metrics and in all 28 UCF-Crime pairs under AP, but in 23 of the 28 under UCF-Crime AUROC. The five exceptions are the five smallest primary-metric gains in the study and are listed beneath Table 6. We attribute the width of these intervals to the resampling unit, since UCF-Crime supplies 290 test videos against 800 on XD-Violence and its AUROC intervals are roughly four times wider on average. Pooled over all 28 pairs, the secondary-metric intervals also exclude zero, at +12.75 [+8.69, +16.95] AP on UCF-Crime and +5.09 [+4.77, +5.41] AUROC on XD-Violence.

A separate paired bootstrap over the smoothed scores backs the smoothing claims of Section 6. The pooled gap interval excludes zero at $\sigma = 4$ and at $\sigma = 8$. Individually, 16 and 17 of the 28 configurations exclude zero at $\sigma = 4$, with seven of the eight binary interfaces among them on each benchmark, and 14 and 16 do so at $\sigma = 8$.

Table 5: Paired video-bootstrap uncertainty for the Yes/No interface (percentage points; 10,000 replicates). Entries are the full-test PROB-GEN gain and the 95% bootstrap interval.

Model	UCF-Crime AUROC	XD-Violence AP
Qwen3-VL-8B	+8.98 [+6.07, +11.61]	+19.59 [+16.46, +22.76]
Qwen2.5-VL-7B	+25.17 [+19.73, +30.13]	+22.39 [+20.16, +24.68]
InternVL3.5-8B	+9.96 [+7.59, +12.56]	+17.99 [+15.36, +20.58]
MiniCPM-V-4.5-8B	+8.38 [+4.86, +11.74]	+18.20 [+15.95, +20.47]
Mean over models	+13.12 [+10.08, +15.88]	+19.54 [+17.61, +21.40]

Scale contrasts within PROB. The resample is shared across scales, so contrasts between scales inside one readout are also paired. At the mean-over-models level, Yes/No minus each of the five finer scales is positive on both benchmarks (+0.56 to +1.64 UCF-Crime AUROC, +2.02 to +3.31 XD-Violence AP), with all five intervals excluding zero on XD-Violence and three of five on UCF-Crime. The per-model ordering is weaker. For the contrast between each model’s best binary and best finer scale, both endpoints are chosen by test-set argmax, so the intervals are conditional on that selection rather than valid for a maximum. Three of four UCF-Crime intervals cover zero, only Qwen3-VL excluding it (+1.05 [+0.11, +2.38]), while on XD-Violence three models exclude zero in favour of binary and Qwen2.5-VL excludes it in the other direction (−2.32 [−3.63, −0.96]). The scale prescription of Section 5.3 is therefore a statement about means, not about every model.

Table 6: Paired video-bootstrap gains by answer scale on each benchmark’s primary metric (percent-age points; 10,000 replicates), with the mean over all 28 model–scale pairs in the last row.

Scale	UCF-Crime AUROC	XD-Violence AP
Yes/No	+13.12 [+10.08, +15.88]	+19.54 [+17.61, +21.40]
0–1 int	+11.73 [+9.25, +14.36]	+17.55 [+15.23, +19.84]
0–5 int	+5.71 [+3.59, +7.87]	+9.38 [+7.39, +11.47]
0–9 int	+5.67 [+4.01, +7.42]	+8.23 [+6.61, +9.89]
0–1 float	+8.78 [+6.05, +11.23]	+8.34 [+7.06, +9.60]
0–5 float	+4.14 [+2.47, +5.89]	+7.12 [+5.53, +8.72]
0–9 float	+4.49 [+2.71, +6.45]	+6.63 [+5.00, +8.25]
All 28 pairs	+7.66 [+5.56, +9.71]	+10.97 [+9.37, +12.58]

Every interval in both tables excludes zero. Five individual UCF-Crime pairs do not, namely MiniCPM-V-4.5-8B at 0–5 int (+2.15 [−0.28, +4.48]) and 0–9 int (+1.83 [−0.00, +3.19]), and Qwen3-VL-8B at 0–1 float (+1.55 [−0.29, +2.98]), 0–5 float (+1.34 [−0.56, +2.75]), and 0–9 float (+0.99 [−0.92, +2.35]). All 28 XD-Violence pairs exclude zero, as do all 28 UCF-Crime pairs under AP.

D PROMPT TEMPLATES

All prompts share the same anomaly semantics and differ only in the rating clause and the required response format. The integer template is instantiated with $M \in \{1, 5, 9\}$ and the floating-point template with $M \in \{1.0, 5.0, 9.0\}$.

Yes/No.

You are reviewing ordered frames from a surveillance video clip.
Is the clip anomalous, suspicious, violent, criminal, or dangerous?
Answer with exactly one word: Yes or No.

Integer scales.

You are reviewing ordered frames from a surveillance video clip.
Rate how anomalous, suspicious, violent, criminal, or dangerous the clip is.
Use a single integer from 0 to M :
0 means completely normal.
 M means clearly anomalous or dangerous.
Answer with exactly one integer from 0 to M and no other text.

Floating-point scales.

You are reviewing ordered frames from a surveillance video clip.
Rate how anomalous, suspicious, violent, criminal, or dangerous the clip is on a scale from 0.0 to M .
0.0 means completely normal.
 M means clearly anomalous or dangerous.
Answer with exactly one float rounded to one decimal place from 0.0 to M and no other text.

E PROMPT SENSITIVITY

The anomaly question is stated in a single wording throughout the paper (Section D), so every reported gap is measured at one point of the prompt space. We therefore rescore UCF-Crime under the Yes/No interface with three paraphrases of that question, for Qwen3-VL-8B and InternVL3.5-8B.

Each paraphrase holds the answer set, the answer polarity, the one-word response constraint, the temporal protocol, and the batch size of the original run fixed and rewords only the framing and the question. A variant asking about normality is deferred to Section G, because reversing the answer polarity changes which answer denotes an anomaly rather than the wording of the question. The base wording is the Yes/No template of Section D, and the three paraphrases are reproduced verbatim below.

pv1: the enumeration of anomaly types replaced by one plain question.

You are reviewing ordered frames from a surveillance video clip.
Is anything unusual or dangerous happening in the clip?
Answer with exactly one word: Yes or No.

pv2: an analyst role, asking about an event rather than the clip.

You are a security analyst monitoring surveillance footage.
Do these ordered frames show an anomalous, suspicious, violent, criminal, or dangerous event?
Answer with exactly one word: Yes or No.

pv3: no second-person framing, the anomaly stated by example.

These are ordered frames from a surveillance video clip.
Does the clip contain an anomaly, such as violence, crime, or a dangerous situation?
Answer with exactly one word: Yes or No.

PROB leads GEN in all eight model–wording cells on both metrics (Table 7). Across the four wordings the gap spans 2.53 and 1.82 AUROC points for the two models and 0.45 and 1.37 AP points, and it never changes sign, staying within 8.55–11.09 and 8.80–10.62 AUROC points. Probability readout is also the steadier readout on the primary UCF-Crime metric, spanning 0.76 and 1.33 AUROC points against GEN’s 2.46 and 1.99, although on AP the two are comparable (2.60 against 2.89, and 1.09 against 0.81). The readout effect is therefore not a property of one phrasing, whereas the absolute values are specific to a wording and should be read as such.

Table 7: Readout gap under three paraphrases of the anomaly question (UCF-Crime, Yes/No, %). *base* is the wording used throughout the paper. Δ is PROB minus GEN.

Model	Prompt	AUROC			AP		
		GEN	PROB	Δ	GEN	PROB	Δ
Qwen3-VL-8B	base	76.28	85.26	+8.98	19.91	40.87	+20.96
	pv1	77.40	85.95	+8.55	19.82	40.83	+21.00
	pv2	74.94	86.03	+11.09	22.72	43.43	+20.71
	pv3	75.96	85.30	+9.34	20.85	41.40	+20.55
InternVL3.5-8B	base	75.97	85.93	+9.96	22.70	39.39	+16.69
	pv1	75.91	84.71	+8.80	22.12	39.60	+17.48
	pv2	73.98	84.60	+10.62	22.41	38.51	+16.11
	pv3	74.55	85.12	+10.58	21.89	38.80	+16.92

F DOES CONSTRAINING THE DECODER CHANGE THE ANSWER?

Section 3.2 treats GEN as an idealized decoder that is always compliant, which is an upper bound on an unconstrained one only if the model would have answered inside the admissible set anyway. Both readouts normalize over the admissible answers and discard the rest of the vocabulary, so the mass the model actually places on that set is not recoverable from a saved run. We therefore rescore UCF-Crime under the Yes/No interface while additionally recording, from the same forward pass, the probability the full vocabulary assigns to $\mathcal{A}_k$ and the unconstrained $\arg \max$ over the vocabulary.

For a single-token answer that $\arg \max$ is the token an unconstrained greedy decoder would emit, so the agreement rate with the constrained decoder is measurable without generating anything.

Over the 69,634 units the mean admissible mass is 0.9999999 for Qwen3-VL-8B and 0.9999750 for InternVL3.5-8B, with minima of 0.9999985 and 0.9999091. For both models the unconstrained greedy token lies inside the answer set on every unit and equals the constrained $\arg \max$ on every unit. Adding the two measurements leaves every reported metric identical to four decimal places, so they are a pure readout of the same forward pass. On this interface, constraining the decoder is therefore not an idealization at all. It selects the answer free generation would have produced, and the parsing failures a real pipeline would add fall on the text around the answer rather than on the answer itself. This is two models on one scale, and a prompt that invited a sentence rather than a single word would not behave this way.

G ANSWER POLARITY

The paraphrases of Section E all ask whether the clip is anomalous, so a model with a standing preference for one of the two answers would bias every cell in the same direction. We therefore rerun UCF-Crime with the question reversed (“Is the clip normal, ordinary, and free of anything suspicious, violent, criminal, or dangerous?”) for Qwen3-VL-8B and InternVL3.5-8B. Reversing the question reverses which answer denotes an anomaly, so the value map is inverted with it and the anomaly score becomes the mass on “No.” Everything else is unchanged. Comparing the two polarities estimates the answer-preference term that a single polarity confounds with the readout.

PROB leads GEN under both polarities on both metrics (Table 8). The readout gap is 7.29 and 9.17 AUROC points under the reversed question against 8.98 and 9.96 under the original, so it is not an artifact of asking about anomaly rather than normality. The answer-preference term itself, PROB under one polarity minus PROB under the other, is 0.20 and 0.09 AUROC points, small enough to ignore on the primary UCF-Crime metric, and 1.85 and 3.76 AP points, which is not negligible and is the reason the two polarities are reported separately rather than averaged.

Table 8: The readout gap is not an artifact of asking about anomaly rather than normality. *anomaly?* is the question used throughout the paper, and *normal?* reverses it and inverts the value map (UCF-Crime, Yes/No, %). Δ is PROB minus GEN.

Model	Question	AUROC			AP		
		GEN	PROB	Δ	GEN	PROB	Δ
Qwen3-VL-8B	anomaly?	76.28	85.26	+8.98	19.91	40.87	+20.96
	normal?	77.77	85.06	+7.29	18.31	39.02	+20.71
InternVL3.5-8B	anomaly?	75.97	85.93	+9.96	22.70	39.39	+16.69
	normal?	76.66	85.84	+9.17	21.74	35.63	+13.89

H PAIRED READOUT GAINS

Figure 7 visualizes the paired PROB–GEN improvement for every model, scale, and dataset, computed on the primary metric of each dataset from Table 3 and Table 4. All 56 cells are positive, and the gains concentrate on the binary scales.

I SCORE CARDINALITY PER MODEL

Table 9 lists the exact number of distinct score values behind Figure 8, for every model, scale, and readout. The most extreme occupancy is MiniCPM-V’s 0–1 floating-point prompt, where a single score covers 94.6% of the units and only four of the eleven admissible values are realized. Averaged over the 28 configurations per dataset, the modal GEN score covers 72.4% of the UCF-Crime units and 55.1% of the XD-Violence units (per-configuration ranges 30.5–96.5% and 31.7–87.8%), and that score is 0.0 in 55 of the 56 configurations. UCF-Crime contains 69,634 scored temporal units and XD-Violence contains 146,449. Under the floating-point scales, PROB realizes 69,538–69,542

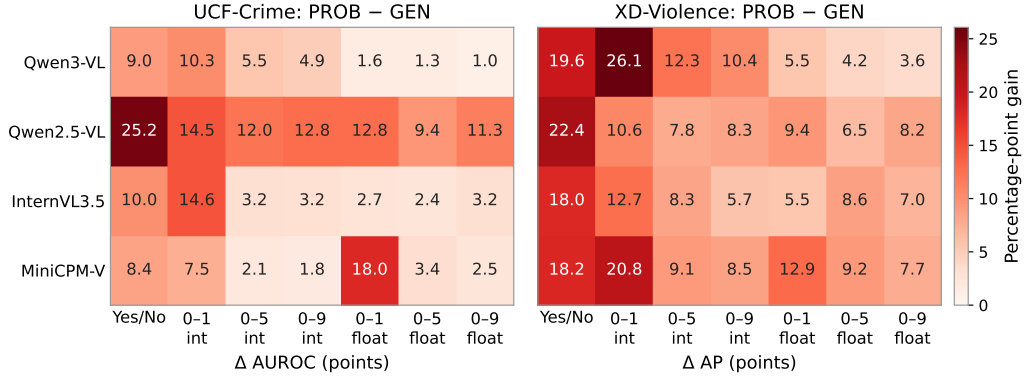


Figure 7: Paired improvement from probability readout. Each cell is PROB minus GEN for the same model, scale, and dataset, with darker marking a larger gain. All 56 cells are positive, and the binary scales gain the most.

distinct values on UCF-Crime and 146,334–146,421 on XD-Violence, a distinct score for more than 99.8% of units. The residual ties come from units whose sampled frames coincide near video boundaries and from numerically identical probabilities, not from the readout. Section T shows that the far smaller counts under the binary scales are set by the arithmetic of bfloat16 inference rather than by the model.

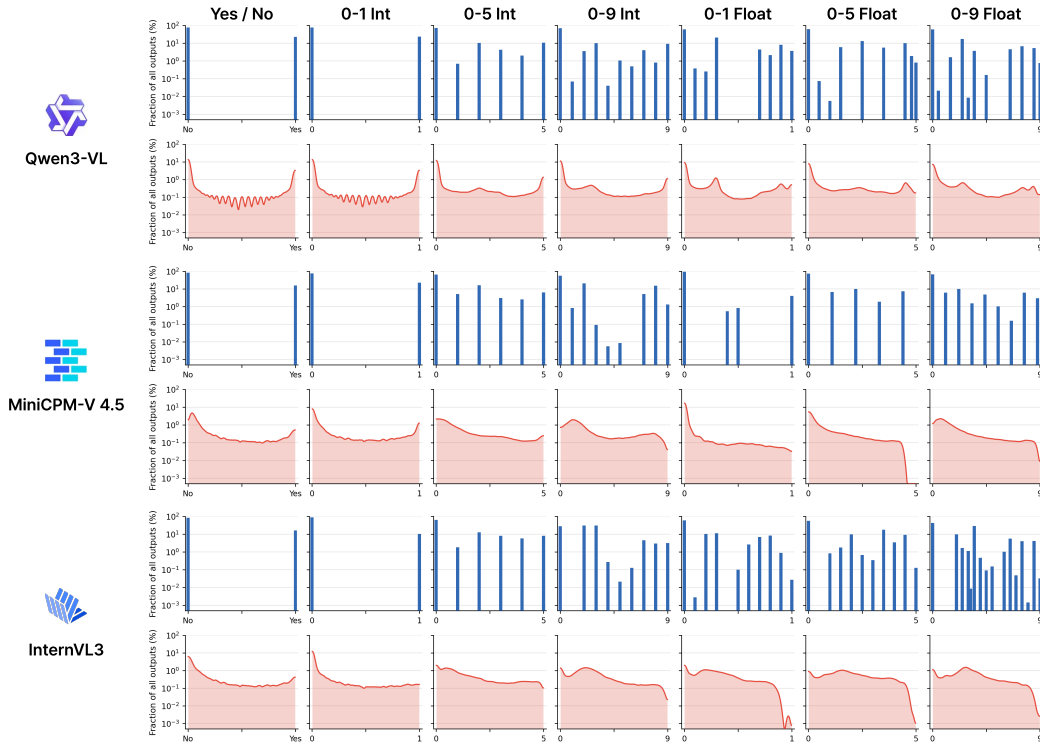


Figure 8: Distinct score values on a log axis, with markers for the four individual VLMs. GEN tracks the nominal level count of the integer scales and realizes at most 18 values even under floating-point prompts that admit up to 91 answers, while PROB approaches one distinct score per temporal unit under the floating-point scales.

Table 9: Distinct score values per model, scale, and readout. Under every floating-point scale, PROB scores more than 99.8% of the temporal units distinctly (69,634 units on UCF-Crime; 146,449 on XD-Violence), while GEN realizes at most 18 values.

Dataset	Model	Yes/No		0–1 int		0–5 int		0–9 int		0–1 float		0–5 float		0–9 float	
		GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB	GEN	PROB
UCF-Crime	Qwen3-VL-8B	2	137	2	273	6	64,353	10	69,409	8	69,538	9	69,542	11	69,542
	Qwen2.5-VL-7B	2	79	2	92	6	30,513	8	65,370	8	69,542	11	69,542	18	69,542
	InternVL3.5-8B	2	133	2	164	6	66,158	9	69,454	10	69,541	10	69,541	16	69,541
	MiniCPM-V-4.5-8B	2	89	2	179	6	34,438	9	64,563	4	69,541	5	69,541	9	69,541
XD-Violence	Qwen3-VL-8B	2	135	2	271	6	126,732	9	146,147	9	146,334	10	146,421	13	146,421
	Qwen2.5-VL-7B	2	82	2	108	5	74,000	8	141,545	9	146,421	17	146,421	18	146,421
	InternVL3.5-8B	2	161	2	186	6	143,261	10	146,388	11	146,420	12	146,421	16	146,421
	MiniCPM-V-4.5-8B	2	108	2	250	6	112,137	10	144,256	8	146,421	6	146,421	10	146,421

J DECOMPOSITION OF THE READOUT GAP

Table 10 lists the exact values behind Figure 3, giving for every scale the mean PROB–GEN gap and its split into tie-breaking (tie-broken GEN minus GEN) and answer reordering (PROB minus tie-broken GEN). Tie-breaking dominates every cell, and the reordering share grows only toward the finest floating-point scales, where the argmax answer and the expected answer value disagree most often.

Breaking the ties at random instead adds 2.30 and 1.67 AP points, 18% and 15% of what PROB recovers. The finer-scale shares quoted in the main text pool the gap and its tie-break component over scales before dividing.

Table 10: Exact decomposition of the PROB–GEN gap (percentage points; mean over the four VLMs). *Tie-break* orders the frames that share a generated score by PROB while keeping the generated scores, and *reorder* is the remaining difference to the full PROB ranking.

Dataset	Scale	AUROC			AP		
		Gap	Tie-break	Reorder	Gap	Tie-break	Reorder
UCF-Crime	Yes/No	13.12	13.12	0.00	18.91	18.91	0.00
	0–1 int	11.73	11.73	0.00	18.11	18.11	0.00
	0–5 int	5.71	5.53	0.18	11.49	10.89	0.60
	0–9 int	5.67	5.41	0.26	11.48	10.17	1.31
	0–1 float	8.78	8.59	0.19	11.00	10.34	0.66
	0–5 float	4.14	3.53	0.62	9.67	8.30	1.38
	0–9 float	4.49	3.29	1.20	8.58	6.12	2.46
XD-Violence	Yes/No	11.00	11.00	0.00	19.54	19.54	0.00
	0–1 int	7.98	7.98	0.00	17.55	17.55	0.00
	0–5 int	3.34	3.22	0.12	9.38	9.17	0.21
	0–9 int	3.06	2.79	0.27	8.23	7.43	0.80
	0–1 float	5.43	5.08	0.35	8.34	7.76	0.58
	0–5 float	2.48	2.03	0.46	7.12	6.65	0.48
	0–9 float	2.31	1.55	0.75	6.63	5.00	1.63

Threshold placement versus compression on the binary scale. A two-valued generated score fixes an operating point as well as a resolution. Its AUROC is determined by the confusion counts of a single implicit threshold, so a badly placed boundary and rank compression both depress it. To separate the two we binarize PROB at the threshold maximizing Youden’s J on the labelled test set, an oracle upper bound on any binary readout of these models that is not attainable without labels. Table 11 reports the split for UCF-Crime under Yes/No. The threshold term ranges from 1.08 to 18.08 points and is essentially Qwen2.5-VL, whose implicit boundary answers “Yes” on only 3.5% of units. The compression floor, PROB minus the oracle binarization, is 7.03–7.60 points for all four models. Compression, not threshold placement, is the model-invariant component of the deficit, and no binarization of these models recovers it.

Table 11: Splitting the Yes/No GEN deficit into threshold placement and compression (UCF-Crime, AUROC, %). Compression, not threshold placement, is the model-invariant part of the deficit. *Oracle bin.* thresholds PROB at the test-set-optimal Youden point, an upper bound on any binary readout. *Threshold* is oracle bin. minus GEN and *compression* is PROB minus oracle bin.

Model	GEN	Oracle bin.	PROB	Threshold	Compression
Qwen3-VL-8B	76.28	77.66	85.26	1.38	7.60
Qwen2.5-VL-7B	60.47	78.55	85.64	18.08	7.09
InternVL3.5-8B	75.97	78.89	85.93	2.92	7.03
MiniCPM-V-4.5-8B	76.07	77.15	84.45	1.08	7.30

Tie-breaking headroom and recovery. Perfect tie-breaking places, among the frames that share a generated score, all anomalous frames first. For AUROC its value equals the tie term of Equation (5), at 15.66 points on UCF-Crime and 9.65 on XD-Violence, averaged over the 28 model-scale pairs. For AP it is worth 36.36 and 29.27 points. The headroom ratios, $2.32\times$ on UCF-Crime and $3.03\times$ on XD-Violence, are what Section 3.4 predicts without reference to any realized gain. The observed gap ratios are smaller, $1.66\times$ and $2.16\times$, so the bound gives the direction of the asymmetry but overstates it by about 40% on both benchmarks.

The discrepancy is a recovery effect, not a failure of the bound. The tie-breaking realized by PROB recovers 7.31 and 4.81 AUROC points, 46.7% and 49.8% of the AUROC bound, but 11.84 and 10.44 AP points, only 32.6% and 35.7% of the AP bound. The same ordering therefore collects roughly 40% less of the headroom under AP than under AUROC, on both benchmarks. This follows from the impurity of the top block. With 34.9% and 58.9% of its frames anomalous, an ordering that improves the block without emptying it of normal frames satisfies AUROC, which only counts pairs, while AP keeps paying precision for every normal frame that remains above an anomalous one. We therefore report the headroom ratio as directional evidence for the mechanism rather than as a quantitative prediction of the gap, since multiplying the bound by the realized recovery share would reproduce the observed ratio (1.62 and 2.17) only because tie-breaking is already 93–95% of the gap (Figure 3), and would test nothing beyond that.

AP sensitivity to tie policy. For each score sequence, we keep the ordering of distinct scores fixed and compute the best and worst item-level AP over all orderings of the frames that share a score. Table 2 averages these values over the 28 model-scale configurations. The grouped value evaluates once per distinct threshold and matches the AP evaluator used throughout the paper. It lies inside the item-level interval but, unlike an arbitrary stable sort, is invariant to the input array order. Generated readout leaves a mean interval wider than 42 points on either benchmark, and probability readout reduces it to roughly one point. The trapezoidal column reports the interpolated estimator of Section O on the same scores, averaged over the same configurations. It falls inside the item-level interval in all 112 score streams. For GEN it exceeds the grouped value in 55 of the 56 streams, by +14.90 and +12.96 points on average. For PROB the offset has no consistent sign, sitting marginally below the grouped value in 20 of the 28 streams on each benchmark, and the positive means of +0.19 and +0.22 come from a few streams that reach +2.59 and +4.42. The remaining PROB interval is concentrated in the binary scales (the widest single configuration spans 9.2 points on UCF-Crime and 17.8 on XD-Violence), while floating-point PROB is almost tie-free. Section T traces these residual ties to the inference arithmetic rather than to the readout.

Ranking quality among frames with the same answer. Table 12 reports the frame-level AUROC of PROB computed only over the frames that share the same generated answer under the Yes/No scale, complementing Figure 5. The frames InternVL3.5-8B answers “No”, the group shown in that figure, are 83.51% of UCF-Crime and 66.61% of XD-Violence. The ordering among the frames answered “No” is strong for every model on both datasets. The single exception is Qwen2.5-VL on XD-Violence, whose frames answered “Yes” are internally close to unordered (46.9). Consistently, this model has the largest overall gap and answers “Yes” least often.

Table 12: Frame-level AUROC of PROB computed only over the frames that share the same generated answer (Yes/No scale).

Model	UCF-Crime		XD-Violence	
	answered “No”	answered “Yes”	answered “No”	answered “Yes”
Qwen3-VL-8B	76.23	71.36	82.43	72.77
Qwen2.5-VL-7B	83.13	60.84	89.38	46.90
InternVL3.5-8B	76.88	69.51	84.22	69.38
MiniCPM-V-4.5-8B	71.33	70.84	83.49	70.00

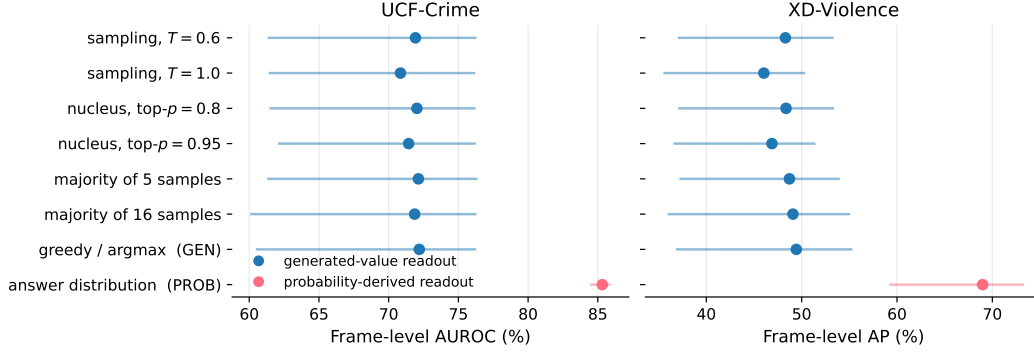


Figure 9: Performance under alternative decoding strategies (Yes/No; mean over four VLMs, bars spanning the models). However the answer is chosen, scoring a unit by that answer spans only 70.85–72.13 UCF-Crime AUROC and 46.03–49.09 XD-Violence AP, which is at or below greedy decoding (72.20, 49.44) and far below PROB (85.32, 68.98).

K AVERAGING SAMPLED ANSWERS

Averaging several decoded answers is the one strategy in Figure 9’s family that leaves the admissible answer set: k samples admit $k+1$ distinct scores. It is also a Monte Carlo estimate of PROB, which is the expected answer value under the same distribution, so the question it raises is not whether it beats GEN but how many decodes it takes to reach what one forward pass already returns. Figure 10 sweeps k from 1 to 256 on the Yes/No interface, averaged over the four VLMs and three seeds.

A single sample ranks *below* greedy decoding, at 70.80 UCF-Crime AUROC and 46.01 XD-Violence AP against GEN’s 72.20 and 49.44. Seventeen available score values are worth less than one well-chosen answer when only one sample supports them, which is the within-answer ordering rather than the score cardinality doing the work. Two samples first exceed GEN. Sixteen recover 72.2% and 59.3% of the GEN–PROB gap, and 256 recover 95.3% and 82.4% while still falling short of PROB. The curve approaches PROB from below and does not cross it, as the estimator relationship requires. Reading the distribution is therefore not only cheaper than sampling it but, at every decoding budget we tested, also better.

L DOES A LARGER MODEL CLOSE THE GAP?

Every VLM in the main study is 7–8B, so the readout gap could in principle be a symptom of limited capacity. We rescore UCF-Crime under the Yes/No interface with the large member of two of those families, Qwen3-VL-32B and InternVL3.5-38B, holding the prompt, the temporal protocol, and both readouts fixed (Table 13).

The gap does not close. It widens by 1.64 AUROC points for Qwen3-VL and 0.91 for InternVL3.5, and by 4.75 AP points for InternVL3.5. The one cell that narrows is Qwen3-VL’s AP gap, by 0.81 points. Capacity is worth 1.58 and 1.54 AUROC points to PROB but -0.06 and $+0.63$ to GEN, so almost the entire benefit of the larger model reaches the ranking through the answer distribution rather than through the answer. Generated readout still realizes exactly two distinct scores at every

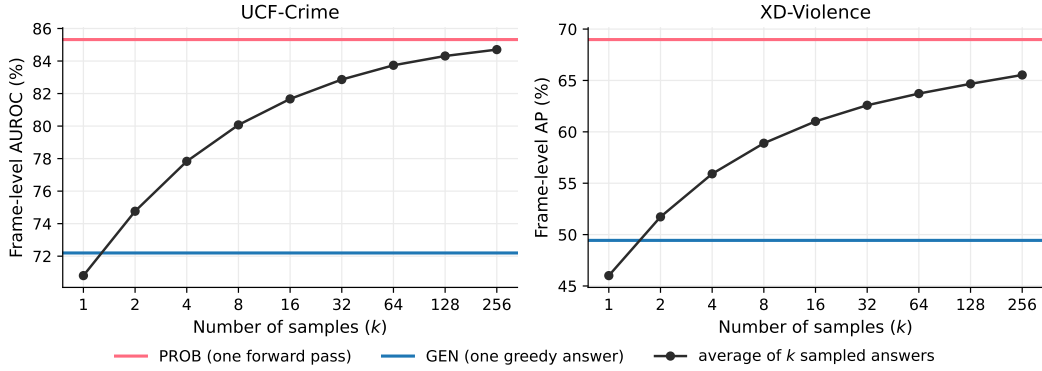


Figure 10: Scoring each unit by the average of k sampled answers (Yes/No; mean over four VLMs and three seeds). The average converges to PROB from below, and one forward pass beats any budget tested.

size, as it must under a binary interface. The number of distinct PROB scores rises from 137 to 288 for Qwen3-VL but stays between 133 and 136 for InternVL3.5, so finer probability resolution is not an automatic consequence of scale. This control covers one benchmark, one scale, and one size step per family, so it is evidence that the gap survives scale rather than a measurement of how it trends with scale.

Table 13: Model-size control on UCF-Crime under the Yes/No interface (percentage points; 69,634 temporal units in every row). *Values* counts distinct realized scores.

Model	GEN		PROB		Δ		GEN/PROB values
	AUROC	AP	AUROC	AP	AUROC	AP	
Qwen3-VL-8B	76.28	19.91	85.26	40.87	+8.98	+20.96	2 / 137
Qwen3-VL-32B	76.22	23.48	86.84	43.63	+10.62	+20.15	2 / 288
InternVL3.5-8B	75.97	22.70	85.93	39.39	+9.96	+16.69	2 / 133
InternVL3.5-38B	76.60	22.43	87.47	43.87	+10.87	+21.44	2 / 136

M WHOSE PROBABILITIES ORDER THE TIED FRAMES?

The ordering hidden behind a fixed answer could in principle be a property of one model’s calibration rather than of the video. We separate the two roles a model plays (emitting the answer that defines the tied block, and ordering the frames inside it) by ranking one model’s tied frames with a different model’s probabilities (Table 14, where the diagonal reproduces Table 12). All 48 off-diagonal cells are above chance, at 71.1–83.5 among frames answered “No” and 64.2–75.4 among frames answered “Yes” on UCF-Crime, 79.2–92.4 and 57.2–75.1 on XD-Violence. Substituting the ranker also leaves the tie-breaking gain positive in every case, between +4.47 and +26.48 AUROC points. An ordering that transfers between independently trained models is evidence about the scene, not an artifact of one model’s probabilities.

The single block that is internally unordered under its own probabilities is recovered by the others. Qwen2.5-VL answers “Yes” on 12.2% of XD-Violence frames and orders them at 46.9, below chance, while the same frames reach 62.9, 63.9 and 68.1 under Qwen3-VL, InternVL3.5 and MiniCPM-V. The evidence is present in those frames, and that model alone fails to express it.

N DETECTION AT A FIXED FALSE-ALARM RATE

AUROC and AP summarize a whole ranking, whereas a deployed detector runs at one threshold. Figure 11 therefore fixes a false-alarm budget and reads off the detection rate, averaged over the

Table 14: Frame-level AUROC among the frames one model ties under the Yes/No interface (rows), ranked by another model’s probabilities (columns). An ordering that transfers between independently trained models is evidence about the scene rather than about one model’s calibration. The diagonal gives the same-model case of Table 12.

Block	Answered by	Ranked by the probabilities of			
		Qwen3-VL	Qwen2.5-VL	InternVL3.5	MiniCPM-V
UCF-Crime					
“No”	Qwen3-VL	76.23	77.23	77.41	71.09
	Qwen2.5-VL	82.58	83.13	83.48	81.53
	InternVL3.5	76.74	77.97	76.88	72.98
	MiniCPM-V	75.86	77.37	77.53	71.33
“Yes”	Qwen3-VL	71.36	71.98	73.28	75.35
	Qwen2.5-VL	69.05	60.84	64.24	64.91
	InternVL3.5	69.14	66.81	69.51	72.22
	MiniCPM-V	69.60	67.99	70.37	70.84
XD-Violence					
“No”	Qwen3-VL	82.43	80.39	80.87	79.19
	Qwen2.5-VL	92.29	89.38	92.37	91.93
	InternVL3.5	86.70	83.93	84.22	83.30
	MiniCPM-V	87.42	84.35	85.72	83.49
“Yes”	Qwen3-VL	72.77	64.58	74.94	75.06
	Qwen2.5-VL	62.95	46.90	63.90	68.09
	InternVL3.5	67.20	57.24	69.38	70.24
	MiniCPM-V	67.54	57.33	70.11	70.00

28 model–scale pairs per dataset. PROB raises it from 8.6 to 16.7 at a false-alarm rate of 0.01 and from 52.0 to 58.1 at 0.1 on UCF-Crime, and from 6.0 to 11.2 and from 54.6 to 66.8 on XD-Violence, winning at least 25 of the 28 pairs at every budget. The middle curve attributes the difference to the ties rather than to a different notion of anomaly, since keeping every generated answer and ordering only the frames that share one already covers 88–97% of the distance. A binary GEN score has no threshold inside this region at all, so the curve drawn for it is the chord across its tied frames, which is what random tie-breaking delivers on average.

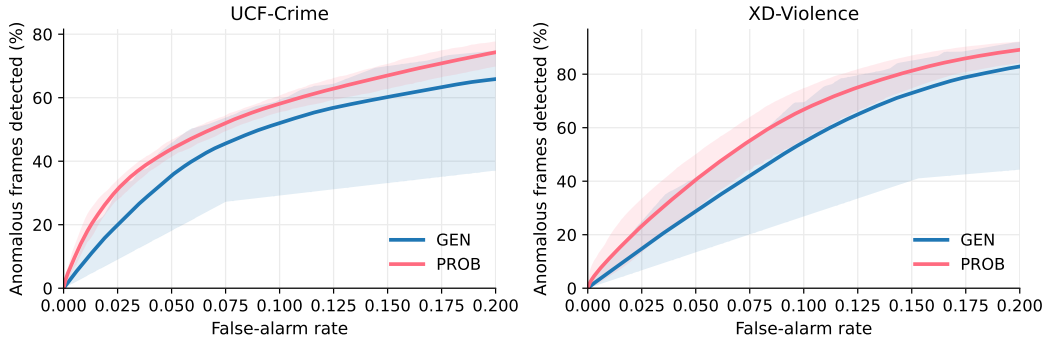


Figure 11: Detection rate against false-alarm rate in the region detectors are operated in, averaged over the 28 model–scale pairs per dataset. Ordering only the frames GEN ties, without changing any answer, covers 88–97% of the distance to PROB.

O WHICH AP ESTIMATOR

Two estimators of average precision are commonly used in VAD. The estimator used throughout this paper evaluates precision at each distinct score threshold and weights it by the resulting recall increment, matching `sklearn.metrics.average_precision_score`. The official XD-

Violence evaluation code instead integrates the precision–recall curve using the trapezoidal rule, a convention also followed by several subsequent VAD implementations. The two estimators agree closely for nearly continuous scores but can diverge substantially when many examples share the same score.

This distinction is consequential for the generated readout. Under GEN, the estimator difference exceeds 20 AP points on the binary scales, whereas under PROB no mean difference exceeds 1.3 points and the discrepancy vanishes as scores become nearly unique. On the binary interfaces, the difference is large enough to reverse the apparent PROB–GEN ordering.

For heavily tied scores, the distinction is substantive rather than merely conventional. Average precision is a weighted mean of the precisions attainable at the score’s own thresholds, with weights that sum to one, so it cannot exceed the largest attainable precision. Equation (6) satisfies that bound in all 112 score streams of the study. The trapezoidal value violates it in 30 of the 56 GEN streams. For InternVL3.5-8B with Yes/No on XD-Violence, no threshold reaches a precision above 59.8%, yet the trapezoidal estimate is 75.25. The mechanism is visible on four frames. Two anomalous and two normal frames sharing one score admit only one threshold, at precision 0.50, but interpolating from the unattainable point (recall 0, precision 1) returns 0.75. The bound is violated in none of the 56 PROB streams, because there is almost nothing to interpolate across.

We do not recompute the published numbers of prior systems. Their scores pass through additional aggregation and temporal post-processing before evaluation, and we do not measure the tie structure of their released score streams. Because the estimator gap depends on that structure, the comparison in ?? reports their published values alongside ours, which are computed with the non-interpolated estimator throughout.

Table 15: Trapezoidal minus non-interpolated AP on the same scores (percentage points; mean over the four VLMs). The gap between the estimators tracks the number of distinct scores, shown for GEN in parentheses.

Scale	UCF-Crime		XD-Violence	
	GEN	PROB	GEN	PROB
Yes/No	+25.99 (2)	+0.45	+20.74 (2)	+0.30
0–1 int	+26.08 (2)	+0.95	+22.27 (2)	+1.25
0–5 int	+15.40 (6)	−0.01	+15.09 (6)	−0.00
0–9 int	+11.20 (9)	−0.01	+9.25 (9)	−0.00
0–1 float	+11.78 (8)	−0.02	+8.98 (9)	−0.00
0–5 float	+9.46 (9)	−0.02	+8.84 (11)	−0.00
0–9 float	+4.40 (14)	−0.01	+5.54 (14)	−0.00

P PER-CLASS RESULTS

Table 16 breaks the Yes/No results down by UCF-Crime anomaly class, evaluating each class’s test videos together with all normal test videos. PROB improves every class on both metrics. The AP gains concentrate on visually salient events, while subtle classes remain difficult under either readout.

XD-Violence annotates each violent video with up to three of six event types, so a video contributes to every class it is labeled with, and each class is again evaluated together with all 300 normal test videos. Table 17 shows the same picture, with every class improving on both metrics and AP gains between +17.0 (Shooting) and +38.6 (Abuse). The contrast between the two Abuse classes is instructive. On UCF-Crime the class has almost no anomalous frames (prevalence 0.02%) and AP stays near zero under either readout, whereas on XD-Violence, at roughly twenty times that prevalence, PROB lifts AP from 10.7 to 49.3. The readout recovers ranking evidence wherever prevalence leaves AP room to express it.

Table 16: Per-class performance on UCF-Crime (Yes/No scale; mean over the four VLMs, percentage points). Every class improves on both metrics.

Class	AUROC			AP		
	GEN	PROB	Δ	GEN	PROB	Δ
Abuse	65.13	94.73	+29.60	0.14	0.33	+0.18
Arrest	85.02	97.48	+12.46	17.45	47.33	+29.89
Arson	67.87	72.92	+5.05	9.27	26.83	+17.56
Assault	89.58	99.33	+9.75	34.15	82.19	+48.04
Burglary	67.08	89.16	+22.08	10.32	32.46	+22.14
Explosion	85.14	95.84	+10.70	15.27	48.30	+33.04
Fighting	87.95	98.52	+10.57	17.81	57.23	+39.42
RoadAccidents	86.87	96.47	+9.60	9.55	29.06	+19.51
Robbery	78.37	96.32	+17.94	10.89	57.42	+46.53
Shooting	77.93	94.50	+16.57	10.40	23.20	+12.79
Shoplifting	56.15	83.29	+27.14	2.25	4.81	+2.56
Stealing	76.16	96.88	+20.72	11.05	45.03	+33.98
Vandalism	78.68	98.16	+19.48	5.58	21.03	+15.45

Table 17: Per-class performance on XD-Violence (Yes/No scale; mean over the four VLMs, percentage points). Every class improves on both metrics. A multi-label video contributes to each of its classes, and every class is evaluated together with all normal test videos.

Class	AUROC			AP		
	GEN	PROB	Δ	GEN	PROB	Δ
Abuse	89.90	99.02	+9.12	10.67	49.28	+38.61
Car accident	76.65	91.68	+15.04	19.37	38.01	+18.64
Explosion	89.81	97.51	+7.70	31.10	54.84	+23.74
Fighting	89.25	96.97	+7.73	45.62	66.35	+20.74
Riot	83.91	96.99	+13.08	60.50	87.44	+26.94
Shooting	89.99	97.11	+7.12	27.30	44.35	+17.05

Q MATCHED SMOOTHING RESULTS

Table 18 reports the mean matched PROB–GEN gap at each smoothing strength, complementing Figure 6. Best- σ selection benefits GEN far more than PROB (6.81 vs. 1.24 UCF-Crime AUROC and 10.42 vs. 2.67 XD-Violence AP points), and on the binary interfaces unsmoothed PROB beats GEN at GEN’s own best σ in 14 of 16 pairs. The mean gap shrinks monotonically as σ grows but remains positive throughout the tested range. At $\sigma=8$, smoothing InternVL3.5-8B’s Yes/No GEN yields 16,391 and 36,426 distinct unit scores on the two benchmarks. Across the grid, PROB is higher in 202 of the 252 matched pairs on UCF-Crime AUROC and in 217 on XD-Violence AP.

Table 18: Mean matched PROB–GEN gap over 28 model–scale pairs at each σ , on both metrics. Values are percentage points, and every column decreases monotonically while staying positive.

σ	UCF-Crime		XD-Violence	
	AUROC	AP	AUROC	AP
0	7.664	12.750	5.086	10.972
1	4.613	7.675	2.453	5.742
2	3.778	6.663	1.816	4.468
4	2.947	5.989	1.382	3.862
8	2.350	4.996	1.015	3.131
10	2.228	4.633	0.904	2.883
16	2.059	3.779	0.692	2.295
20	1.976	3.392	0.609	2.049
24	1.916	3.193	0.552	1.869

R JOINT SCORING AND EXPLANATION

Across the full grid of 16 cells, two metrics, and nine smoothing settings, PROB exceeds GEN in all 288 comparisons. Table 19 gives the full primary-metric results for the joint scoring–explanation experiment. The prompt requests one concise, factual description of the visible scene in addition to the score, with the response constrained to two labeled lines. *Score-first* emits the score before the description and *explanation-first* reverses that order. Descriptions are decoded greedily up to 64 tokens, and both readouts are evaluated at the score slot exactly as in Section 3.

Table 19: PROB exceeds GEN in every cell of the joint scoring–explanation interfaces (percentage points, with Base the score-only interface). *S-first* decodes the score before the explanation, and *E-first* decodes the explanation first.

Model	Scale	Base		S-first		E-first	
		GEN	PROB	GEN	PROB	GEN	PROB
<i>UCF-Crime, AUROC</i>							
Qwen3-VL-8B	Yes/No	76.28	<u>85.26</u>	76.08	<u>85.07</u>	75.12	<u>84.46</u>
Qwen3-VL-8B	0–1 int	75.23	<u>85.49</u>	76.56	<u>84.96</u>	75.18	<u>84.37</u>
MiniCPM-V-4.5-8B	Yes/No	76.07	<u>84.45</u>	74.64	<u>82.34</u>	75.72	<u>83.38</u>
MiniCPM-V-4.5-8B	0–1 int	76.60	<u>84.10</u>	76.07	<u>83.39</u>	74.20	<u>81.94</u>
<i>XD-Violence, AP</i>							
Qwen3-VL-8B	Yes/No	50.66	<u>70.25</u>	51.11	<u>71.50</u>	48.33	<u>65.03</u>
Qwen3-VL-8B	0–1 int	43.35	<u>69.42</u>	47.67	<u>64.44</u>	47.56	<u>58.86</u>
MiniCPM-V-4.5-8B	Yes/No	54.97	<u>73.17</u>	53.96	<u>73.18</u>	53.26	<u>70.92</u>
MiniCPM-V-4.5-8B	0–1 int	51.01	<u>71.84</u>	48.81	<u>71.90</u>	51.30	<u>67.91</u>

Effect of output order. Under score-first prompting, PROB differs from its score-only counterpart by -0.90 points on average, with five of eight cells within one point. Under explanation-first prompting, PROB falls below the score-only baseline in every cell, by -3.39 points on average and up to -10.57 AP, while GEN changes by only -0.44 points on average. On a two-answer scale the decomposition of Section 5.2 degenerates. GEN is the indicator $\mathbb{I}[\tilde{p}_t(\text{Yes}) > \frac{1}{2}]$, a monotone coarsening of PROB, so ordering the GEN blocks and breaking their ties by PROB reconstructs the PROB order exactly and the reordering term is zero by construction, not by measurement. The 24 runs of this study confirm the implementation realizes the identity (maximum absolute difference 0.00 on both metrics) rather than corroborating the 92–95% share, which is an average over scales where the split can fail. What an explanation costs is therefore measurable as a loss of tie-breaking quality.

Table 20 takes three measurements at the score slot. The generated answer is largely preserved, since explanation-first decoding changes it on 4.5–9.6% of scored units relative to the score-only interface (mean 6.7%). The distribution behind that answer is not preserved. Mean entropy of the two-answer distribution falls from 0.219 to 0.127 bits, a 42% reduction that holds in all eight cells (23–59% per cell), whereas score-first decoding leaves it at 0.208 bits and lowers it in only six of eight, the exceptions being the two MiniCPM-V-4.5-8B 0–1 cells. Writing α for the probability that PROB orders correctly a positive–negative frame pair whose generated scores are equal, and $\Pr(s^+ = s^-)$ for the frequency of such pairs, Equation (5) gives $\text{AUROC}_{\text{PROB}} = \text{AUROC}_{\text{GEN}} + \Pr(s^+ = s^-)(\alpha - \frac{1}{2})$, so $\alpha = 50\%$ is chance. Averaged over the eight cells, α reads 74.6% at Base, 73.8% under score-first decoding (lower in five of eight, the three that rise all on XD-Violence) and 72.5% under explanation-first decoding (lower in seven of eight, the exception being MiniCPM-V-4.5-8B with Yes/No on XD-Violence, 75.7% to 77.8%). These drops of 0.8 and 2.2 points track the PROB changes of -0.90 and -3.39 points, and a compressed distribution breaking fewer of the remaining ties correctly is what the main text means by evidence compressed behind an unchanged label. The largest score-first drop (Qwen3-VL-8B with the 0–1 scale on XD-Violence, -4.98 AP) is also where PROB’s own scores saturate. The probability that PROB itself leaves a positive–negative pair tied rises from 1.91% at Base to 8.53% under score-first and 11.82% under explanation-first decoding, the finite-precision effect noted in Section 6.

Table 20: Explanation-first decoding compresses the answer distribution and breaks fewer ties correctly. Score-slot measurements are the mean entropy H (bits), the share α of tied pairs PROB orders correctly (chance 50), and the units whose answer differs from Base under E-first.

Model	Scale	Base		S-first		E-first		
		H	α (%)	H	α (%)	H	α (%)	Δ ans. (%)
<i>UCF-Crime</i>								
Qwen3-VL-8B	Yes/No	0.117	74.5	0.073	74.3	0.071	73.9	6.2
Qwen3-VL-8B	0–1 int	0.118	77.1	0.085	73.4	0.050	73.7	7.0
MiniCPM-V-4.5-8B	Yes/No	0.423	71.2	0.316	68.3	0.265	69.7	4.7
MiniCPM-V-4.5-8B	0–1 int	0.321	70.7	0.547	70.0	0.222	69.0	9.6
<i>XD-Violence</i>								
Qwen3-VL-8B	Yes/No	0.095	74.1	0.059	75.2	0.074	72.7	4.5
Qwen3-VL-8B	0–1 int	0.100	79.4	0.069	74.2	0.047	68.5	8.6
MiniCPM-V-4.5-8B	Yes/No	0.360	75.7	0.213	80.0	0.149	77.8	4.8
MiniCPM-V-4.5-8B	0–1 int	0.218	74.5	0.305	74.9	0.136	74.4	8.2
Mean		0.219	74.6	0.208	73.8	0.127	72.5	6.7

S FLOATING-POINT TOKEN ACCOUNTING

For Qwen3-VL, Qwen2.5-VL, and InternVL3.5, the floating-point evaluation records three token-accounting conventions: full numeric sequence, decimal-point-cost removal, and digit-only aggregation. The MiniCPM-V-4.5 runs record the full numeric sequence only. The GEN score is identical across all three conventions in every run. Across scales and smoothing values, the maximum difference in the primary PROB metric is 0.00018 percentage points on UCF-Crime (AUROC) and 0.0081 points on XD-Violence (AP). All reported results therefore use the full-sequence likelihood.

T INFERENCE PRECISION

The small number of distinct PROB values that Table 9 reports under the binary scales is a property of the inference arithmetic and not of the model’s uncertainty. A two-candidate PROB score is a function of the difference between the two answer log-likelihoods, and in bfloat16 the logits carry an eight-bit significand, so that difference is confined to multiples of the significand step at their magnitude. On UCF-Crime with Qwen3-VL-8B and the Yes/No prompt the step is 0.25 in log-odds, and 121 of the 137 realized values lie within 10^{-3} of a multiple of it, against 0.80% expected by chance, with a median distance to the grid below 10^{-7} . The 16 exceptions all have log-odds above 10, where the stored probability is within a few float32 units of one and recovering the log-odds from it loses precision. Softmax saturation is not the mechanism, since only nine of the 69,634 units reach the largest probability the readout represents ($1 - 2^{-23}$).

Rescoring the same configuration with float32 weights, leaving the prompt, temporal protocol, and both readouts unchanged, raises the number of distinct PROB values from 137 to 65,885. The grid disappears, with 0.85% of the values near a multiple of 0.25 against 0.80% expected by chance. The AP tie interval narrows from 2.06 to 0.26 points, the latter being the floor set by broadcasting one score over the 16 frames of a temporal unit. The ranking is otherwise unaffected, since the two runs agree at Spearman 0.9985 over the 69,634 units and AUROC moves from 85.26 to 85.27. The runs also differ in batch size, so we do not attribute their remaining metric differences to precision alone. All reported results use bfloat16, the standard inference setting. What the coarser arithmetic puts at risk is the tie interval rather than the metric value.

U ADDITIONAL LIMITATIONS

The evidence base is narrow. The evidence base is four open 7–8B VLMs, two surveillance benchmarks, and one temporal protocol. Other model families, domains, and temporal units may exhibit different effect sizes, and the joint-explanation, prompt-sensitivity, polarity, and inference-

precision controls each cover a subset of that grid, the last a single cell. Five of the 28 UCF-Crime pairs, all small gains on fine scales, have bootstrap intervals including zero (Section C).

Unconstrained generation is out of scope. The admissible-mass control (Section F) covers two models on one scale, and no interface we test speaks to a system prompted to answer freely, which could answer outside the admissible set and add parsing failures around the answer.

The scale axis is confounded. The per-model ordering among scales is unresolved, with the exceptions quantified in Section C. More fundamentally, changing the number of levels also changes the label wording and its tokenization, so scale is not a clean granularity axis. Isolating granularity would require scales that hold the verbalizer and its tokenization fixed while varying only the number of levels, which the prompt formats we use cannot do.

Why the retained order exists is not measured. Ranking evidence does not settle why order survives inside a fixed answer. One possible explanation is that the answer likelihood varies continuously with visual evidence while the argmax reads it through a fixed decision boundary. Two results are consistent with that reading, the cross-model transfer of within-answer order (Section M) and the entropy drop under explanation-first decoding that lowers PROB while barely moving GEN (Table 20). Testing it directly would require controlled perturbations of the visual input while tracking the answer likelihood, which our ranking protocol does not provide.

V CONTEXT AMONG EXISTING VAD SYSTEMS

For context, Table 21 and Table 22 place a single fixed interface, InternVL3.5-8B with the Yes/No prompt, among published VAD systems. This comparison does not establish a new state of the art. The listed pipelines differ in visual context, temporal design, and post-processing, and their published values are reported by their own authors under their own AP estimators. The exhibited interface is the running example of Section 5, fixed before this comparison, and $\sigma = 4$ is the single smoothing value applied to both readouts on both datasets. Within our own grid, stronger PROB configurations exist on UCF-Crime (86.46 AUROC, Table 3), while on XD-Violence this cell is the grid’s maximum (73.31 AP against a next-highest 73.17, Table 4). Its purpose is to express the readout effect on a familiar scale. With GEN, the model trails LAVAD (Zanella et al., 2024) on UCF-Crime AUROC and, under the non-interpolated estimator of Equation (6), on XD-Violence AP. That second half is estimator-dependent, since the same GEN scores yield 75.25 AP trapezoidally against 55.32 non-interpolated. PROB applied to the same forward pass exceeds the strongest training-free entry (85.93 vs. 84.74 AUROC on UCF-Crime, 73.31 vs. 68.85 AP on XD-Violence) and moves by less than half a point between the two estimators. Under a single matched smoothing setting ($\sigma = 4$ on both datasets), both readouts improve, yet smoothed GEN (83.38 AUROC, 67.65 AP) remains below even unsmoothed PROB, while smoothed PROB reaches 86.73 AUROC and 75.46 AP. The choice of readout is thus comparable in magnitude to recent progress among training-free systems.

Table 21: Comparison with prior VAD systems on UCF-Crime (frame-level AUROC and AP, %). The bottom rows apply the two readouts of Section 3 to the same frozen InternVL3.5-8B with the same Yes/No prompt, without smoothing and with one matched smoothing setting ($\sigma = 4$); the readout alone separates the two unsmoothed rows by 9.96 AUROC points. Bold marks the best training-free value in each column.

Method	Supervised	AUC (%)	AP (%)
RTFM (Tian et al., 2021)	Weakly Supervised	84.03	–
MSL (Li et al., 2022)	Weakly Supervised	85.62	–
S3R (Wu et al., 2022)	Weakly Supervised	85.99	–
MGFN (Chen et al., 2022)	Weakly Supervised	86.98	–
FDPN (Song et al., 2025)	Weakly Supervised	88.03	–
LLaVA-1.5 (Liu et al., 2024)	Training Free	72.84	–
Video-Llama2 (Zhang et al., 2023)	Training Free	74.42	–
LAVAD (Zanella et al., 2024)	Training Free	78.33	–
EventVAD (Shao et al., 2025)	Training Free	82.03	–
VADTree (Li et al., 2026)	Training Free	84.74	–
InternVL3.5-8B, GEN (Yes/No)	Training Free	75.97	22.70
InternVL3.5-8B, GEN (Yes/No, $\sigma=4$)	Training Free	83.38	31.14
InternVL3.5-8B, PROB (Yes/No)	Training Free	85.93	39.39
InternVL3.5-8B, PROB (Yes/No, $\sigma=4$)	Training Free	86.73	40.95

Table 22: Comparison with prior VAD systems on XD-Violence (frame-level AUROC and AP, %). The bottom rows use the same fixed interface as Table 21 (InternVL3.5-8B, Yes/No), without smoothing and with the same matched smoothing setting ($\sigma = 4$); the readout alone separates the two unsmoothed rows by 17.99 AP points. Bold marks the best training-free value in each column.

Method	Supervised	AUC (%)	AP (%)
RTFM (Tian et al., 2021)	Weakly Supervised	–	77.81
MSL (Li et al., 2022)	Weakly Supervised	–	78.58
S3R (Wu et al., 2022)	Weakly Supervised	–	80.26
MGFN (Chen et al., 2022)	Weakly Supervised	–	80.11
LLaVA-1.5 (Liu et al., 2024)	Training Free	79.62	50.26
Video-Llama2 (Zhang et al., 2023)	Training Free	80.21	53.57
LAVAD (Zanella et al., 2024)	Training Free	82.89	60.02
EventVAD (Shao et al., 2025)	Training Free	87.51	64.04
VADTree (Li et al., 2026)	Training Free	90.55	68.85
InternVL3.5-8B, GEN (Yes/No)	Training Free	85.31	55.32
InternVL3.5-8B, GEN (Yes/No, $\sigma=4$)	Training Free	91.03	67.65
InternVL3.5-8B, PROB (Yes/No)	Training Free	91.67	73.31
InternVL3.5-8B, PROB (Yes/No, $\sigma=4$)	Training Free	92.38	75.46